Fintech Sentiment Intelligence Analysis

Executive Summary

Project Duration: 2 weeks
Dataset: 10,386 customer reviews
Apps Analyzed: Cash App, Chime, PayPal, Venmo
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🎯 Objective

Analyze customer sentiment across 4 major fintech apps to: 1. Identify hidden pain points and crisis language patterns 2. Build an AI-powered severity detection system 3. Provide competitive intelligence and actionable insights

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📊 Key Achievements

Model Performance Improvements

- **+87% increase** in severity detection accuracy (macro-F1: 0.16 → 0.30)
- **+17% increase** in high-severity issue detection (F1: 0.47 → 0.55)
- **-6% reduction** in negation-related errors

Business Value Delivered

- **707 hidden negatives identified** (complaints disguised as positive reviews)
- **Crisis keyword lexicon** built with 50+ fintech-specific terms
- **Interactive dashboard** enabling real-time competitive analysis
- **200 reviews hand-labeled** for validation ground truth

🔍 Critical Findings

1. Account Access Issues Drive Severity

35% of high-severity complaints involve locked/frozen accounts

- Keywords: "frozen", "locked", "can't access", "suspended"
- Average severity: **4.2/5**
- Recommendation: Prioritize account access team & improve unlock processes

2. Hidden Negatives in Positive Reviews

707 reviews (2.8%) rated 4-5★ but contain severe complaints

- Example: "Love this app! But my account has been frozen for 3 weeks..."
- Risk: Traditional sentiment analysis misses these entirely
- Solution: Our severity-based detection flags them automatically

3. Competitive Landscape

App	Avg Rating	Reviews Analyzed	Negative Reviews
Cash App	4.07★	2,600	~30%
Chime	3.89★	2,595	~30%
PayPal	3.43★	2,594	~30%
Venmo	3.31★	2,597	~30%

Insight: Cash App leads in customer satisfaction (4.07★); Venmo has lowest rating (3.31★). Dataset: 10,386 total reviews with 30.6% negative (rating ≤ 2)

🛠️ Technical Approach

Data Pipeline

1. **Scraped 10,400 reviews** from Google Play Store (2,600 per app)
2. **Cleaned & normalized** text (removed 14 duplicates, fixed encoding) → 10,386 final
3. **Built sentiment engine** using VADER with custom enhancements
4. **Validated model** on 200 hand-labeled reviews

AI Enhancements

- **Negation Handling**: Pre-processes "not good" → correctly classified as negative
- **Crisis Lexicon**: 50+ fintech keywords (fraud, frozen, hacked, dispute)
- **Severity Scoring**: 1-5 algorithm combining sentiment intensity + crisis keywords + star rating
- **Hidden Negative Detection**: Flags positive/neutral sentiment with high severity

Tech Stack

- **Languages**: Python 3.9+
- **NLP**: VADER, NLTK, spaCy
- **Data**: Pandas, NumPy
- **Visualization**: Streamlit, Plotly, Seaborn

📈 Business Impact

For Product Teams

✅ **Prioritize features** based on severity-weighted complaints
✅ **Track trends** over time to measure improvement
✅ **Compare** against competitors for positioning

For Customer Support

✅ **Auto-flag critical cases** (severity 4-5) for immediate escalation
✅ **Identify hidden complaints** missed by traditional filters
✅ **Reduce response time** for urgent money/access issues

For Marketing

✅ **Benchmark sentiment** against competitors
✅ **Craft messaging** addressing top pain points
✅ **Highlight strengths** relative to competition

🚀 Deliverables

1. ✅ **Interactive Dashboard** (4 pages, 15+ charts)
2. ✅ **Sentiment Engine** (`src/analysis.py` - production-ready)
3. ✅ **Validation Report** (200-review ground truth)
4. ✅ **Crisis Keyword Lexicon** (50+ terms)
5. ✅ **GitHub Repository** (fully documented)

🔮 Recommended Next Steps

Short-term (1-3 months)

- Deploy sentiment engine as **REST API** for real-time monitoring
- Set up **automated daily scraping** for trend tracking
- Integrate with **support ticketing system** for auto-escalation

Medium-term (3-6 months)

- Expand to **iOS App Store** reviews
- Add **Spanish language support** (large fintech demographic)
- Build **predictive churn model** based on review patterns

Long-term (6-12 months)

- Implement **BERT-based sentiment** for higher accuracy
- Add **aspect-based analysis** (what specific features are complained about)
- Create **executive dashboard** with weekly trend alerts

💼 Skills Demonstrated

✅ **Data Engineering**: Web scraping, ETL pipelines, data cleaning
✅ **NLP & Machine Learning**: Sentiment analysis, text classification, model validation
✅ **Product Analytics**: User research, competitive analysis, insight generation
✅ **Data Visualization**: Interactive dashboards, storytelling with data
✅ **Software Engineering**: Clean code, documentation, version control

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